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Middle-Distance Running

Jean Claude Vollmer and Martin Buchheit

Performance Demands of Middle-Distance Running

In this section, we introduce track and field's 1500 m event and discuss the various factors of importance and the relative contribution that physical performance, running technique, and tactics make toward winning races. Note that the data presented and analyzed throughout the chapter was mostly available from male athletes, and slight differences may be apparent when comparing across genders. The chapter is written predominantly from the point of view and experience of coach Jean Claude Vollmer, with the scientific interpretation of Martin Buchheit.

Sport Description and Factors of Winning

In the sport of track and field, or athletics, the middle-distance (MD) events refer to distances run over 800 to 2000 m on an athletic track, including the 1500 m and 1 mile races, which are run at relatively high speeds (table 13.1). Middle-distance events differ from long-distance races (3000 m, 5000 m, 10,000 m), which must be run at slower speeds (i.e., 24.5 to 22.82 km/h for 3000 to 10,000 m WR speeds, respectively). It is important to note however that the average speed of the 1500 m is only 2.3 km/h slower

than that of the 800 m, despite being almost twice as long, while it is 1.9 and 2.4 km/h faster than the 3000 m and 5000 m races, respectively. In terms of running demands, the 1500 m event is a race that sits right in the middle between short, very high speed races (800 and 1000 m), which require high muscle power and strength, and longer endurance events (3000 and 5000 m), which require more muscular endurance. This explains the related observation, whereby, in addition to runners with a specific 1500 m profile, the 1500 m event features both 800 m and 3000/5000 m profile athletes, who can all be successful. Thus, determining a runner's locomotor profile is a compulsory practice for MD coaches to

Table 13.1 Middle-Distance Events, World Record Times, and Associated Running Speeds

Event	World record time (min:sec)	Speed (km/h)
800 m	1:40:91	28.54
1000 m	2:11:96	27.28
1500 m	3:26:00	26.21
Mile	3:43:13	25.28
2000 m	4:44:79	25.28

be able to highlight strengths and weaknesses and consequently determine training priorities.

My analysis (personal observation, J.C. Vollmer) of 800 m to 5000 m race performances of the top 51 all-time 1500 m performers (as of July 21, 2017, range: 3:26.00 world record to 3:30.77) is rich in lessons. These athletes' performances over distances in the ballpark of the 1500 m event are revealing with respect to each athlete's likely phenotypic makeup. For example, we can take performances achieved by these 51 athletes over specific distances ranging from 800 to 5000 m, so long as an athlete is within 98% of the mean fastest distance-specific run time. Doing so provides us with the following results:

- 800 m (n = 44): mean 1:45:18, range 1:41:73 to 1:49:86
- 1000 m (n = 40): mean 2:16:07, range 2:12:18 to 2:20:33
- Mile (n = 49): mean 3:48:81, range 3:43:31 to 3:55:62
- 2000 m (n = 24): mean 4:55:1, range 4:44:79 to 5:06:34
- 3000 m (n = 41): mean 7:39:0, range 7:23:09 to 8:08:08
- 5000 m (n = 32): mean 13:18:95, range 12:50:24 to 14:19:19

From this data, we can divide these short-, middle-, and long-distance specialists into three respective groups:

- Group 1: 17 runners, best performers over 800 m, 1000 m, and 1500 m
- Group 2: 18 runners, best performers exclusively over 1500 m
- Group 3: 16 runners, best performers over 1500 m, 2000 m, 3000 m, and 5000 m

Note that the distances of 1000 m, 2000 m, and 3000 m are not official competition distances in championships, although the 3000 m is an official indoor distance. The merging of the runners into these three groups shows how the 1500 m event is a race distance of energetic convergence, bringing together runners with very different locomotor profiles.

The first profile group (800, 1000, and 1500 m, n = 17) (figure 13.1) had an average 1500 m time of 3:29:57. Athletes show the following performances: 800 m (n = 17) 1:43:5, 1000 m (n = 17) 2:14:3, mile (n = 17) 3:48:5, 2000 m (n = 8) 4:54:3, 3000 m (n = 13) 7:47:7, 5000 m (n = 7) 13:31:9. These runners display excellent anaerobic capabilities and speed qualities (all runners able to run at around 46.5 over 400 m), which allows them to be in the finals and on podiums in big event championships over 800 m. If they can still perform well over 2000 m, these runners rarely venture beyond their favorite shorter distances. The upper limits of the longer distances include Cram in 4:51:3 over 2000 m, Ngeny in 7:35:5 over 3000 m, and Ramzi with 13:10:7 in the 5000 m. The only one to present a notable performance over 10,000 m is Webb at 27:34:7. Extremely powerful over 1000 m, their performance is typically limited over 3000 m.

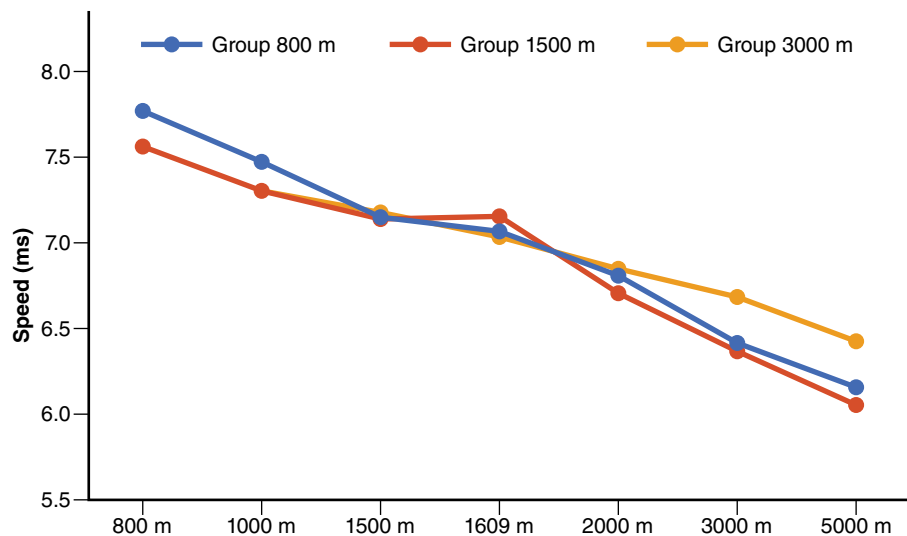


Figure 13.1 Running speed over the different race distances. Error bars omitted for clarity.

Table 13.2 Winners of Major Championships Since 1980 Appearing in the World Top 50 as a Function of Their Locomotor Group Profile

Group 800 m (n=8 athletes for 12 titles)	Group 1500 m (n=3 athletes for 3 titles)	Group 3000 to 5000 m (n=3 athletes for 10 titles)
Coe (Olympics 1980 and 1984) Cram (Worlds 1983) Bile (Worlds 1987) Ngeny (Olympics 2000) Ramzi (Worlds 2005) Kiprop (Olympics 2008, Worlds 2011, 2013, 2015) Kamel (Worlds 2009) Makloui (Olympics 2012)	Rono (Olympics 1988) Cacho (Olympics 1992) Centrowitz (Olympics 2016)	Morceli (Worlds 1991, 1993, 1995, Olympics 1996) El Guerrouj (Worlds 1997, 1999, 2001, 2003, Olympics 2004) Lagat (Worlds 2007)

Previous championships (Olympics only) won by Walker (group 1, 1976), Vasala (group 1, 1972), Keino (group 3, 1968), Snell (group 1, 1964), Elliott (group 1, 1960), Delany (group 1, 1956), Barthel (group 1, 1952), and Eriksson (group 2, 1948).

The second profile group (1500 m, $n = 18$) (figure 13.1) had an average 1500 m time of 3:29:9. Athletes show the following performances: 800 m ($n = 16$) 1:45:6, 1000 m ($n = 14$) 2:17:3, mile ($n = 17$) 3:44:7, 5000 m ($n = 9$) 13:46:0. This group represents athletes who can perform really well in only the 1500 m. While they are capable of adequate performances over 800 m, they cannot reach podiums. For the most part, they also do not perform well in the upper distances (apart from Cacho, who ran 7:32:9 over 3000 m). They are true 1500 m distance specialists who have developed the relevant physiological capacities and motor skills.

The third profile group (1500, 2000, 3000, 5000 m, $n = 16$) (figure 13.1) had an average 1500 m time of 3:28:9. Athletes show the following performances: 1000 m ($n = 10$) 2:17:1, mile ($n = 17$) 3:48:8, 2000 m ($n = 11$) 4:51:7, 3000 m ($n = 14$) 7:29:0, 5000 m ($n = 17$) 12:58:5. Equipped with exceptional aerobic capacity, this group can also challenge on the world stage over 5000 m. They reach a limit at the 10,000 meters (except for Farah, who developed into a 5000 m to 10,000 m runner). Their great aerobic potential allows them to finish quickly on the last lap.

It should be noted that only one runner, Saïd Aouita (5000 m Olympic champion in Los Angeles), who is a 1500 m (3:29:5), 2000 m (4:50:8), 3000 m (7:29:5), 2 mile (8:13:5), and 5000 m (12:58:0) world record holder, could also claim a group 1 position, with a 1:43:3 over 800 m. Unique in history, he won the bronze in 800 m in Seoul, but got injured and withdrew from the 1500 meters.

Interestingly, the short- and long-distance group profiles tended to have better results in the 1500 m

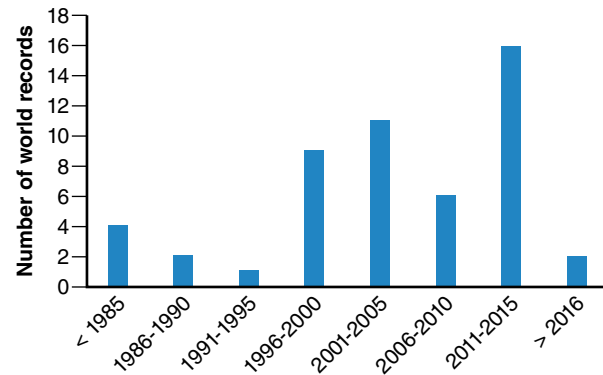


Figure 13.2 Time course of the 51 world best performances in the 1500 m, as of July 21, 2017. Note that this data set included 51 races, where the 50th and 51st performances elicited the exact same time.

event than the 1500 m specialist group profiles in terms of personal bests and podiums at major competitions (table 13.2). As well, there were five runners in group 3, three in group 2, and two in group 1 in the world top 10, as well as ten in group 3, six in group 1, and four in group 3 in the world top 20 for 1500 m.

Note that runners were already running pretty fast over 1500 m more than 30 years ago. The 50 best performances ($n = 51$) ranging from 3:26:0 (world record) to 3:30:77 were achieved at different times (figure 13.2). Comparing the periods of achievement of the top 50 best performances in the various disciplines, it can be seen that while for the 800 m event, seven performances were achieved before 1989, there were no top 50 performances prior to 1996 in the

5000 and 10,000 m events. This shows that disciplines with a strong aerobic dominance continued to progress more than the races relying on more mixed energetic profiles. Among the 51 best performances ever in the 1500 m, 23 were performed in Monaco; 7 in Brussels; 4 in Zürich; 3 in Berlin, Rome, and Paris; 2 in Rieti and Nice; and 1 in Cologne, Doha, Lausanne, and Seville. The Monaco meeting has become the fastest meeting in history, and even more in recent years. Among the 18 best performances of the top 51 that were performed since 2012, 17 were achieved in Monaco on the Louis II Stadium track.

Relative Contribution of Physical Performance

As outlined, the metabolic and neuromuscular underpinnings of a 1500 m runner are essential, and even the phenotype engine makeup of the individual may be a performance determinant (i.e., groups 1 to 3). However, tactics also play a major role in performance outcome, as this impacts how the engine may actually be used during a race.

Championship Race: Twelve competitors will qualify to start during a final of a world championship or Olympic Games. Historically, this number

decreased from 12 to 9 at the Melbourne Olympics (10 at the Munich Olympics) before returning to 12 since the Helsinki World Championships in 1983. The number of runners on the starting line influences a key aspect of this event: the tactics, which are an important determinant of success (figure 13.3). If the race objective was essentially to run as fast as possible, as it is for a record race, we might see hares launch during a race to help achieve this objective. During a championship race, it's a completely different story. The 12 runners on the starting line of a major finals all have the will to win, bringing 12 different strategies or racing tactics to the fore, which evolve during the race according to the circumstances. Occurrences might include starting quickly in the first 100 m; running at a constant pace "on the train" (i.e., one after the other) at a sustained high speed; using intermittent accelerations, a progressive acceleration, a sudden surge, or sometimes even teamwork tactics as each country can have three runners start if they manage to qualify during the preliminary rounds. Thus, the position of a runner in the peloton at key moments during the race is essential, especially toward the end of the race. These aspects all need to be addressed and simulated in situational training sessions to prepare the athlete.

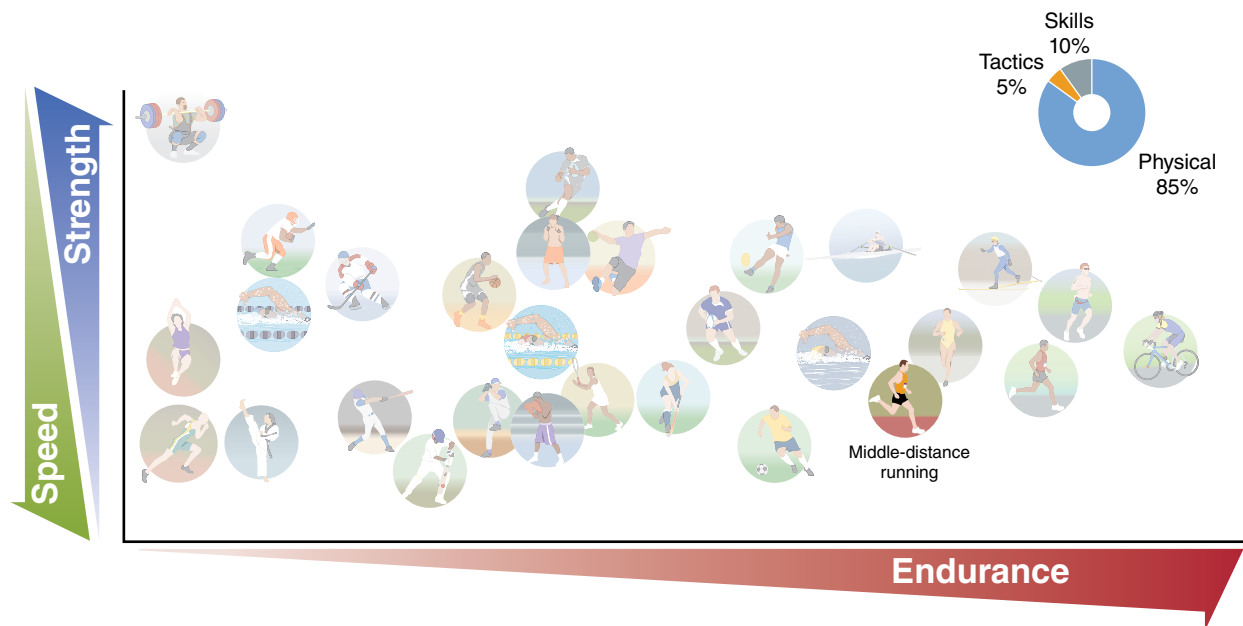


Figure 13.3 The position of the runner on the three axes illustrates the relative importance of the three main physical capacities for elite middle-distance 1500 m competition. Outside of critical psychological aspects, which are difficult to quantify, the pie chart shows the general relative importance of physical competence (85%), skills (10%), and tactical awareness (5%) for success.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

Another aspect that can't be neglected comes from the impact of successive races occurring before the final in qualifying rounds (series, semifinals, finals, interspersed by a rest day). Like a Tour de France road cyclist who must perform day after day, the MD runner must have an aerobic system that is highly engaged to facilitate recovery between rounds. Since the end of the 1980s, few runners double across the 800 m and 1500 m male competitions. Exceptions include Coe in Los Angeles (winner in 1500 m and second in the 800 m) and P. Elliott in Seoul in 1988 (fourth in the 800 m and second in the 1500 m). While runners more often double across 5000 m and 10,000 m (they have one week of recovery between events), few have ever doubled across 1500 m and 5000 m, with the exception of El Guerrouj in Athens. Most recently, however, Makloufi completed the 800 m and 1500 m double successfully (2×2 nd) in 2016, which is remarkable considering the difficulty of the sequencing of the races, inducing a high density of work in a very short period of time at the highest level.

To gain further insight into the role that pacing and tactics play in 1500 m performance outcomes, we analyzed performance results of the major championships of the modern era (28 Olympic finals and World Championships since the Munich Olympics in 1972). Here, we found the average time of the winner was 3:35:55 (± 4.10), with performance times ranging from 3:27:66 (Hicham El Guerrouj) at the 1999 World Championships in Seville to Matthew Centrowitz's victory at the Rio Olympics in 3:50:01 (the worst winning time in an Olympics final since 1932!). This extremely large (22.45 s) time difference between winners demonstrates the large role that tactics play in 1500 m race outcomes (figure 13.3). For example, in a study of world mile records, Noakes et al. (12) showed that the slowest lap occurred in 90% of the races in either the second (34%) or third (56%) laps. The last lap was only ever slowest in two races (6%), while in 24 races (76%) it was either the fastest (38%) or second fastest (38%). Record races are special events generally organized for a single runner, in which the only goal is to break the record time. These types of races do not have the richness of a championship, with its wavering pacing strategies and twists and turns. As shown by Thiel (16) in his study of MD events at the Beijing Olympic Games, the profile of a championship race is completely different, including random occurrences and a pace marked by microvariations throughout, even when pace might appear regular to the casual

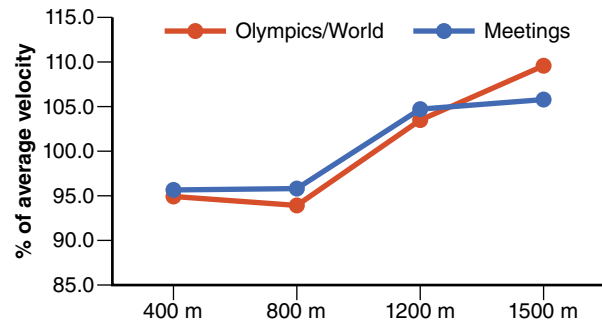


Figure 13.4 Comparison of the pacing profiles during Olympics and World Championships (in which runners chase medals) versus meetings or record races (in which runners try to break records).

observer. The only consistent occurrence is that the gold medal always is won at the end by a sprint over the last 400 m. Our analysis of the 28 finals of the 1500 m at Olympics and World Championships since 1972 shows the pacing pattern illustrated in figure 13.4, alongside the comparison profile of a typical meeting or record race.

Only at the World Championships in Seville (1999) and the Sydney Olympics (2000) did these typical patterns (figure 13.4) not occur. At those races, the first 400 m was markedly faster than the second and third laps due to team tactics; the pace was pushed to the benefit of Hicham El Guerrouj.

Typically, the faster the first 800 m, the slower the last 400 m due to energetic aspects. Thus, fatigue resistance is a particularly important element to train for in the 1500 m event to meet two important requirements:

1. being able to respond to a rapid first 800 m start without burning out metabolically speaking, and
2. being able to finish quickly over the last lap of the race whatever the race situation (table 13.3).

In general, the race is decided across the last 100 m of the 1500 m race, irrespective of the preceding antics.

Nevertheless, over 1500 m, runners can lose a race because of too fast a start or inappropriate accelerations applied at certain moments of the race to gain positions, but what is clear is that 1500 m races are won predominantly in the second part of the race. To win, exceptional fatigue resistance and a high top speed are essential characteristics of all podium athletes.

It should be noted also that the last 200 m of a 1500 m event are often run faster than the last 200 m

Table 13.3 Average, Slowest, and Fastest Running Times (With Average Speed) Measured Over Select Running Portions of the World Top 50, 1500 m Event Performances

Lap	Average time (s)	Slowest time (s)	Fastest time (s)	Average speed (m/s)
Last 800 m	1:50:15 ± 1:33	1:54:07 (Sydney Olympics 2000)	1:46:78 (Athens Olympics 2004)	7:27
Last 500 m	1:06:42 ± 1:24	1:08:93 (Edmonton 2001)	1:04:24 (Stuttgart 1993)	7:58
Last 400 m	52:30 ± 1:24	54:6 (Edmonton 2001)	50:46 (Barcelona 1992)	7:69
Last 300 m	39:30 ± 1:18	41:18 (Edmonton 2001)	37:57 (Barcelona 1992)	7:69
Last 200 m	26:50 ± 0:70	28:03 (Edmonton 2001)	25:18 (Montreal 1976)	7:69

of the 800 m event. Thus, the aerobic aspect is vital in the 1500 m to preserve the important anaerobic resources and neuromuscular freshness. Put another way, an athlete's aerobic foundation spares his speed reserve so it can be released on command.

Conversely, when racing starts too fast (over the first 400 m), which is often the case in record races, achieving a good result also requires high levels of fatigue resistance toward the end of the race. Clearly, the 1500 m event is a complex race requiring both exceptional aerobic capabilities and also top anaerobic abilities (basic speed and fatigue resistance). The 1500 m runner is therefore a particularly complete runner, which is why the 1500 m may be the most interesting middle-distance running race and is often termed a blue-ribbon event in athletics.

Targets of Physical Performance in Middle-Distance Running

Studies have shown that the contribution of aerobic energy supply to a 1500 m simulated race ranges from 77% (5) to 84% ± 1% for well-trained male runners (14) and about 86% for women (5). Note that even from these comparisons across genders, where it's expected that males would have a greater degree of muscularity due to the influence of higher basal testosterone levels, that an individual's degree of muscularity appears related to his or her anaerobic capabilities (18). $\dot{V}O_2\text{max}$ values of 1500 m runners have been reported as 72 ± 2 ml·kg⁻¹·min⁻¹ for well-trained males (5), 75.1 ± 6.1 ml·kg⁻¹·min⁻¹ for national-level runners (1) and 78.1 ± 3.7 ml·kg⁻¹·min⁻¹ ($v\dot{V}O_2\text{max}$: 20.44 ± 1 km/h) for elite French runners (6). Two of the best French runners ever, Baala (3:28:98, 12th fastest performance of all time;

1:43:15 for 800 m; two gold medals at European Championships; 2003 silver medalist; Beijing Olympic bronze medalist) and Bob Tahri (3000 m steeplechase specialist; 3:32:79 for 1500 m; World bronze medalist in 2009; two-time medal winner at the European Championships; two-time Olympic finalist; six-time finalist in the world from 2003 to 2011; retained European record in the 3000 m steeplechase in 8:06:91 and then 8:01:18) displayed $\dot{V}O_2\text{max}$ values of 79.11 ml·kg⁻¹·min⁻¹ and 81.49 ml·kg⁻¹·min⁻¹, respectively.

Interestingly, Boileau et al. (2) showed $\dot{V}O_2\text{max}$ to be more correlated with performance for middle distance (68.9 ml·kg⁻¹·min⁻¹, $r=0.70$) versus long-distance runners (76.9 ml·kg⁻¹·min⁻¹, $r=0.32$), suggesting the importance of this physiological variable for predicting 1500 m event success. This may be related to the fact that $\dot{V}O_2\text{max}$ would be reached during the event approximately 78 ± 17 s after the start, corresponding to a distance of 464 ± 67 m in runners finishing a 1500 m race in 4:08 ± 9 (7, 8). Once $\dot{V}O_2\text{max}$ is achieved, $\dot{V}O_2$ generally drops correspondingly with the reduction in running speed before stabilizing during the middle portion of the race and then clearly dropping during the last 300 m; in fact, the final acceleration relies strongly on the anaerobic metabolic contribution.

Despite Tahri's very high $\dot{V}O_2\text{max}$ (81.49 ml·kg⁻¹·min⁻¹) and exceptional performances from 3000 m to 10,000 meters, he only ever presented very good, but not top, performance over 1500 m (3:32:79). Tahri's profile confirms that to be a world top performer over 1500 m, qualities other than $\dot{V}O_2\text{max}$ are required. In fact, Ingham et al. (10) showed, using an allometric model, that the maximal speed associated with $\dot{V}O_2$, in direct relation to running economy, can explain 95.9% of the performance variance. Anaerobic capacity is

also a determinant, as evidenced by the relatively high blood lactate values measured post race (18.4 mmol/L^{-1}) following a 1500 m event (11). Other measures of anaerobic work capacity and neuromuscular power, such as the Margaria stair climbing test and vertical jump power, respectively, have also been shown as being related to 1500 m event outcome (9).

As a consequence, 1500 m runners have a very complete physiological and locomotor profile, as

described by top coaches including Peter Coe (4), the father and coach of S. Coe (former 800 m and 1500 m world record holder) and H. Wilson (19), coach of S. Ovett (former world record holder of the 1500 m) as well as from academic studies (3, 13, 15, 17). A nonexhaustive list of the required qualities for successful 1500 m participation are provided in detail in table 13.4 and illustrated simply in figure 13.5.

Table 13.4 Typical Characteristics of Successful 1500 m Runners

Height	The longer the event, the shorter the runners (18) • 800 m: $182.5 \pm 6.2 \text{ cm}$ • 1500 m: $179.8 \pm 5.2 \text{ cm}$ • 3000 m steeplechase: $178.2 \pm 4.2 \text{ cm}$ • 5000 m: $175.5 \pm 2.6 \text{ cm}$ • 10,000 m: $174.8 \pm 3.2 \text{ cm}$
Body mass	The longer the distance, the lighter the runners (18) • 800 m: $67.6 \pm 4.8 \text{ kg}$ • 1500 m: $64.8 \pm 4.9 \text{ kg}$ • 3000 m: $65.0 \pm 4.8 \text{ kg}$ • 5000 m: $61.5 \pm 4.7 \text{ kg}$ • 10,000 m: $60.3 \pm 4.6 \text{ kg}$
Metabolic aspects	• A high $\dot{V}\text{O}_2\text{max}$ and, most importantly, a high $\dot{V}\text{O}_2\text{max}$ (linked to running economy) • A high anaerobic threshold (the endurance factor, 1500 m runners are also capable of performances over 5000 m and 10,000 m). The aerobic factor is essential because it allows the runner to release his speed.
Neuromuscular aspects	• A good race economy, which influences $\dot{V}\text{O}_2\text{max}$ • A high anaerobic capacity and power • High lower-limb neuromuscular power (leg strength and power, foot quality) • A good level of basic speed with the ability to accelerate • Good running technique and lower-limb efficiency
Tactical intelligence	1500 m runners must have good awareness of their capacities in relation to the race situation, and be able to adapt their plan accordingly to maximize their outcome.
Strong ability to tolerate pain	Every race inevitably elicits very high blood lactate levels (and associated acidosis) (11).
Age	1500 m runners are relatively young when compared to runners in other events. The average age of the top 50 all-time 1500 m runners is 24.8 ± 3.0 years. Excluding those runners still active who are likely to beat their best performances in the future, the average becomes 25.2 ± 3.2 yr. This average age has decreased slightly compared to a study conducted in 1995 (average age = 25.5 yr). This difference can be explained by the arrival of Kenyan runners at earlier maturity but likely similar training ages. In general, Caucasian runners tend to mature later, between 2 to 3 yr later on average. The breakdown by age group is as follows for the world top 50: • <20 yr: 3 • 20 to 23 yr: 12 • 23 to 26 yr: 14 • 26 to 29 yr: 15 • 29 to 32 yr: 5 • >32 yr: 2 The oldest top performer was William Tanui (33 years old). This average age is slightly higher than that of 800 m runners (24.2 ± 3.2 yr) but less than that of 3000 m steeplechase runners (25.9 ± 2.8 yr), 5000 m runners (26.2 ± 3.7 yr), and 10,000 m runners (26.4 ± 3.9 yr).

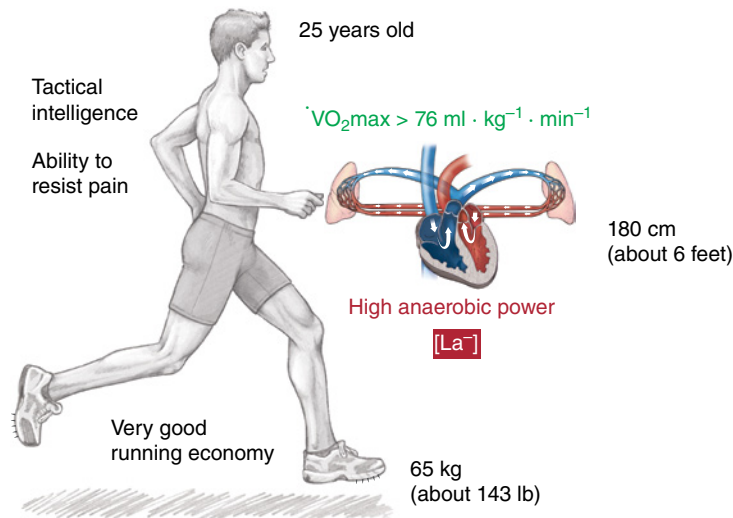


Figure 13.5 Ideal profile of a top 1500 m performer.

Key Weapons, Manipulations, and Surveillance Tools

Recall that weapons refer to the high-intensity interval training (HIIT) formats we can use to target the physiological responses of importance. In this section, we present the different HIIT weapons used within a 1500 m runner's training program, their manipulations, and ways to monitor their effectiveness (surveillance).

HIIT Weapons

The distribution of training content (% of total weekly volume) for an elite 1500 m runner is as follows:

- Aerobic work carried out as both continuous (submaximal) and in intermittent (HIIT) forms represents 80% (winter period) to 70% of the training content depending on the season,
- Race and speed technique work represents 5% to 10%,
- General physical preparation and strength work represents 5% to 10%,
- Work around the specific race pace represents 0% (winter period) to 15%, and
- Miscellaneous work (e.g., other sporting activities such as cycling, swimming, cross-country skiing, mountain biking, etc., to add variety) represents 0% to 5%.

With a greater volume of metabolic conditioning completed as continuous submaximal running (80%

to 90%) and with the intensity determined, depending on the time of year and athletes, using percentages of $\dot{V}O_{2\max}$, $HR_{\max}$, lactate threshold, race-specific speed or even RPE, HIIT sessions become the key and nonnegotiable components of training used to improve race performance, despite making up only 10% to 20% of all training sessions combined. As detailed throughout the book, the majority of practitioners, if not all, use different forms of HIIT, which vary in terms of the intensity or format. To prepare for the 1500 m, as with other distances, HIIT is used for three main reasons (or targets) when it comes to:

1. Improving oxidative function, namely ($\dot{V}O_{2\max}$ (type 1, 2, 3, 4 targets),
2. Improving anaerobic function, namely anaerobic power and capacity (type 3 and 4 targets), and
3. Targeting and developing responses and adaptive comfort at race-specific speed (type 4 targets), with specific emphasis on achieving adaptations at the neuromuscular level.

At different times of the preparation, and depending on the athlete's profile, I (JCV) will use many forms of HIIT as either short (50%) or long intervals (50%; figure 13.6), with the respective proportion between the two varying according to the training phases. Runs last from 10 s to 8 min (i.e., time to exhaustion at $\dot{V}O_{2\max}$), using repetitions from 80 to 3000 m at an intensity ranging from 90% of $\dot{V}O_{2\max}$ (3000 m) to almost maximal (10 s).

Number of repetitions (3 to 20) and series become a direct function of the work interval duration and session objectives.

Manipulations of Interval Training Variables

The attainment of specific HIIT responses, or hitting select target types, for each individual runner, at



Figure 13.6 Percentage of the different HIIT formats (weapons) used throughout the annual season for elite 1500 m runners. Note here that medium-length intervals (1-2 min) are used often for middle-distance runners, referred to as long intervals consistently throughout the rest of the book.

key moments in preparation, is achieved via the manipulation of at least one of the 12 variables presented in chapter 4 (figure 4.2). Figure 13.7 illustrates this in a 1500 m runner, using a 300 m repetition example, in which adjustment of the recovery characteristics (and only this) allows us to reach three distinct targets.

Continuing with the 300 m repetition example using sessions designed for a top 1500 m runner (PB: 3:36 s, $\dot{V}O_{2\max}$ 22.5 to 22.8 km/h), HIIT target types (figure 1.5) can additionally be manipulated using several variables at a time, such as the intensity of the work interval and recovery characteristics (table 13.5). Given the importance of running at the right intensity, and in turn, targeting the right acute biological responses, controlling running speed is paramount. It is important not to start too fast in these HIIT sessions, as doing so creates excessive fatigue that can compromise both the quality of the training and the ability of the athlete to complete session repetitions. Several strategies can be used to ensure that runners run at the desired intensity, including:

- Run one or two straight lines (100 m) at the end of the warm-up at the target pace,
- Give split times (using whistle or voice) during rehearsals,

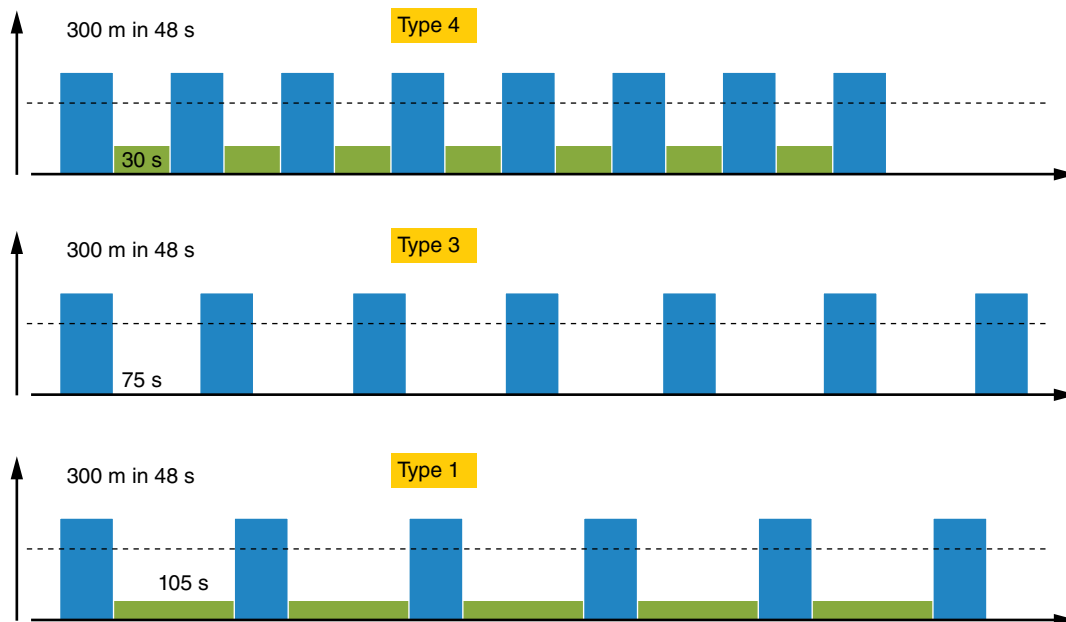


Figure 13.7 Manipulation of recovery interval characteristics and its effect on HIIT responses. Recovery = 30 s at 50% $\dot{V}O_{2\max}$ (type 4); recovery = 75 s passive (type 3); recovery = 105 s easy jog (type 1).

- Use an external pacing beeper to give the athlete the right rhythm, or have the athlete set the splits on his watch (although this may require reprogramming or readjustment if the coach changes the session during training),
- Coach uses bike to pace runner appropriately, and
- Use of other experienced runners as pacers.

Additionally, specific HIIT target-type example sessions I've (JCV) used with my athletes are shown in tables 13.6 to 13.10.

HIIT with long intervals typically run 94% to 96% of $\dot{V}O_{2\max}$ across durations ranging from 2:30 to 8:00 (and possibly up to 10:00) or distances from 800 m to 3000 m. This work can be done both outdoors (forest, soccer pitch, the preferred option) or on a track. HIIT with moderate-duration intervals typically run 98% to 104% of $\dot{V}O_{2\max}$ over 1 to 2 min duration. This work is mostly done on the track. HIIT with short intervals can be run 105% to 108% of $\dot{V}O_{2\max}$ over durations of less than 1 min, and are systematically executed on the track.

Table 13.5 HIIT Variable Manipulation Effects on Target Types Using 300 m Intervals

HIIT target	Session	Work intensity	Recovery duration	Recovery modality	Between series
Type 1	15×300 m	95% $\dot{V}O_{2\max}$ (49 to 50 s)	45 s	Jog	
Type 2	2×(6×300 m)	105% $\dot{V}O_{2\max}$ (45.5 to 46 s)	1 min, 30 s	Jog	6 min walk or jog
Type 3	8×300 m	115% $\dot{V}O_{2\max}$ (41 to 41.5 s)	3 min	Walk and jog	
Type 4	6×300 m	120% $\dot{V}O_{2\max}$ (38.5 to 39 s)	8 min	Walk	
Type 5	4×300 m	Near max speed (36 s)	12 min	Sitting or walking	

Table 13.6 Example of Type 1 HIIT Sessions Used to Prepare for a 1500 m Race

Session	Work intensity	Recovery duration	Recovery modality	Between series
2×(3×1000 m)	95% $\dot{V}O_{2\max}$	2 min	Jog	4 min jog
4×1500 m	92% $\dot{V}O_{2\max}$	2 min, 30 s	Jog	
12×400 m	98% $\dot{V}O_{2\max}$	1 min	Jog	

Table 13.7 Example of Type 2 HIIT Sessions Used to Prepare for a 1500 m Race

Session	Work intensity	Recovery duration	Recovery modality	Between series
Fartlek (30 to 45 min)	Near $\dot{V}O_{2\max}$ (RPE) for 45 s to 3 min, outdoors for a total of 15 to 20 min	30 s to 1 min, 45 s	Jog	
2×(300/400/500/600/500/400/300 m)	100% $\dot{V}O_{2\max}$ for 45 s to 1 min, 30 s	45 s to 1 min, 15 s	Jog	6 min walk or jog

Table 13.8 Example of Type 3 HIIT Sessions Used to Prepare for a 1500 m Race

Session	Work intensity	Recovery duration	Recovery modality	Between series
2 (6×300 m)	110% $\dot{V}O_{2\max}$	2 min, 30 s	Walk, jog	6 min walk
3 (3×500 m)	105% $\dot{V}O_{2\max}$	1 min, 30 s	Jog	4 min walk

Table 13.9 Example of Type 4 HIIT Sessions Used to Prepare for a 1500 m Race

Session	Work Intensity	Recovery duration	Recovery modality	Between series
3 × (800 m + 400 m)	800 m at 1500 m speed 400 m at 800 m speed	3 min	Jog	8 min walk
5 × (400 m + 200 m)	400 m at 1500 m speed 200 m at increasing speed	None		6 min walk
2 (4 × 300 m)	>1500 m speed	3 min	Walk	8 min walk

Table 13.10 Example of Type 5 HIIT Sessions Used to Prepare for a 1500 m Race

Session	Work intensity	Recovery duration	Recovery modality
500 m + 400 m + 300 m	800 m speed	10 to 12 min	Walk
3 × 500 m	>1500 m speed	6 min	Walk
4 × 400 m	>1500 m speed	3 min, then 2 min, then 1 min	Walk

Load Surveillance and Monitoring Tools

To quantify training load in my runners, I tend to count repetitions at different specific speeds (e.g., race pace, $\dot{V}O_2\text{max}$) and look at cumulated training time and distance run across each macrocycle. I also often ask my runners for their post-session RPE as a way of judging session intensity, although I don't use RPE as part of a global load measure.

Training Status Surveillance and Monitoring Tools

While nothing will ever replace the coach's eye, the main training status monitoring tool I use is the stopwatch. The monitoring of laps or split times during repetitions (average times but also progression of these split times throughout the session) is very useful for gauging progress and detecting a fatigued state. I also measure lactate and HR responses in certain HIIT sessions (i.e., tables 13.8 to 13.10) and especially HR drift between HIIT repetitions, as well as post-session HR recovery as indices of both fitness and freshness. Along these same lines, but for longer run types only, I observe the GPS-speed/HR relationship as a proxy of running economy.

The standardized HIIT sessions, generally performed indoors to avoid environmental fluctuations and associated reduced reliability, might include:

- Post aerobic cycle training assessment (every 8 weeks): 4 × 1000 m at $\dot{V}O_2\text{max}$ (interspersed

with 1 min, 45 s of passive rest with lactate measured after 1 min) + 5 min passive recovery + all-out 400 m (lactate sample after 3, 10, and 15 min).

- Race pace target session: 4 × 1000 m at 1500 m speed (interspersed with 5 min passive rest with lactate measured after 3 min) + all-out 500 m (lactate sample after 3, 10, and 15 min).
- Anaerobic capacity on a treadmill: 40 s at $\dot{V}O_2\text{max}$ on a 10% grade + 15 min passive rest with lactate measured after 3, 10, and 15 min + run to exhaustion at $\dot{V}O_2\text{max}$ on a 10% gradient. I use the difference between the post-40 s lactate and post-run-to-exhaustion lactate as a measure of anaerobic capacity.

Strategies for Structuring the Training Program Using HIIT

The implementation of HIIT within the overall program is as much a science as it is an art. Here, I use a blend of experience, combined with an understanding of physiology, biology, and an athlete's history and profile to come up with what I believe is the ideal HIIT option that will best develop the athlete appropriately at any particular time in the program.

Controllable and Uncontrollable Factors

I have mostly always worked with professional athletes, whose lives and daily duties have been mostly

planned around their training times. For this reason, we rarely need to compromise training for work or other commitments. This factor makes training the priority and shapes the athlete’s life around the most important training sessions to permit optimal recovery. Following these guidelines and, in particular, having these programmed an especially long time ahead of the main competitions provides good control over the training cycles and daily content. While you can always be caught off guard by last-minute details (e.g., family, traffic, media obligations) that can derail the process, the programming for MD runners can typically be well implemented.

Program Structure and Progression

I generally organize my yearly plan using five typical macrocycles:

- 1. Preparatory phase/development phase 1 (October to March)
- 2. Preparatory phase/development phase 2 (March to mid-May)
- 3. Precompetition (mid-May to end of June)
- 4. Competition (July to August)
- 5. Taper (before main competition)

Each of these cycles has different aims in terms of their physiological adaptation goals and performance development, with the quantity and type of HIIT sequences varying accordingly (table 13.11).

Incorporation of HIIT

HIIT sequences are generally part of a complete training session, which also include warm-up drills (running technique, often before HIIT) and sometimes strength and coordination exercise (before or after). The form of these sessions often influences the training contents and sequences, e.g., running drills before an HIIT sequence are more likely to be performed on a track or the grass part of the track than in a forest.

Sample Training Programs

In this section, I present several examples of the typical HIIT weapons that I program within each macrocycle (table 13.11) to target each specific HIIT type for two 1500 m runners who display the contrasting locomotor profiles described in the first part of the chapter, namely a group 1 versus group 3 individual (figure 13.1).

Preparatory Period (October to Mid-May)

The preparatory period is the longest overall period of the year and lasts from October to mid-May for most runners without indoor competition.

The 20% of HIIT training time in phase 1 sees a distribution as follows: type 1, 15% (table 13.12); type 2, 3% (table 13.13); type 3, 2% (table 13.14); type 4, 0%; type 5, 0%. This breakdown is detailed in tables 13.12 through 13.14. Note: There are no

Table 13.11 Typical HIIT Target Type Emphases Throughout the Season for Elite 1500 m Runners

Macrocycle	% of HIIT HIIT sessions/total weekly sessions	Type 1	Type 2	Type 3	Type 4	Type 5
Preparatory phase\development phase 1 October to March	20% (2 or 3 HIIT/12 sessions)	15%	3%	2%		
Preparatory phase\development phase 2 March to mid-May	25% (3 HIIT/12 sessions)	10%	5%	5%	5%	
Precompetition mid-May to end of June	20% (2 or 3 HIIT/12 sessions)			5%	10%	5%
Competition June to August	15% (2 HIIT/12 sessions)		5%		5%	5%
Taper (before main competition)	10% (2 HIIT/12 sessions)				5%	5%

Table 13.12 HIIT Type 1 (15%) in Developmental Phase 1 of the Preparatory Period in Elite 1500 m Runners With Group 1 Versus Group 3 Profiles

HIIT with long intervals at 92% to 96% of $\dot{V}O_{2\max}$ for effort times from 2 min 30 s to 8 min (and possibly up to 10 min) or distances from 800 m to 3000 m. This work can be done both outdoors (forest, soccer pitch) or on a track.	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
• 4 × 1200 m in steps of 400 m to 92%/94%/96% of $\dot{V}O_{2\max}$ with a jog recovery of 3 min • Fartlek as 3 min, 4 min, 2 min, 2 min, 4 min, 3 min (pace between 92% to 94% of $\dot{V}O_{2\max}$) with jogged recoveries of 1 to 2 min (duration and pace of sensations) • Fartlek as 2 min, 3 min, 2 min, 4 min, 3 min, 2 min, 3 min, 2 min (pace between 94% and 96% $\dot{V}O_{2\max}$) with recoveries of 1 min to 1 min, 30 s of active jog (duration and pace by feel)	• 3 × 6 min or 3 × 2000 m at 95% $\dot{V}O_{2\max}$ with active recovery of 1:30 between sets and 3 min active recovery between series • 2 (3 × 3 min) or 2 (3 × 1000 m) to 95 or 96% of $\dot{V}O_{2\max}$ with active recovery of 1:15 and 4 min between series • 3000 m/2000 m/1000 m at 92%/94%/96% of $\dot{V}O_{2\max}$ with active recovery of 3 min and then 2 min • 1000 m/2000 m/2000 m/1000 m at 96%/95% of $\dot{V}O_{2\max}$ with 2 min, 3 min, 2 min jog recovery
HIIT with moderate-duration intervals (duration of 1 to 2 min effort) from 98% to 102% of $\dot{V}O_{2\max}$.	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
• 3 (400/600/400 m) at 100%/102%/102% $\dot{V}O_{2\max}$ with passive recovery of 90 s between reps and 4 min passive between series • 2 (5 × 400 m) at 102% of the $\dot{V}O_{2\max}$ with passive recovery of 1:30 between reps and 4 min passive recovery between series • 3 (600+500+400 m) at 100%/102% of $\dot{V}O_{2\max}$ with 1:30 and then 1:15 of passive recovery between reps and 4 min passive between series	• 3 (2 × 800 m) to 98%/102% of $\dot{V}O_{2\max}$ with active recovery of 1:45 and 4 min between series • 4 (2 × 600 m) of 100% to 102% of $\dot{V}O_{2\max}$ with recovery of 1:30 jog and 3 min between series • 2 (4 × 500 m) from 100% to 102% $\dot{V}O_{2\max}$ with recovery of 1:15 jog and 5 min between series
HIIT with short intervals (duration of effort less than 1 min) from 104% to 106 % of $\dot{V}O_{2\max}$	
Both profiles	
• 2 (6 × 300 m) at 104% $\dot{V}O_{2\max}$ with light jog recovery and 3 min passive between series • 3 (6 × 200 m) at 106% $\dot{V}O_{2\max}$ with 1:00 to 1:15 jog recovery and 6 min passive between series • 4 (200/300/200 m) at 106%/104%/106% $\dot{V}O_{2\max}$ with 1:45 jog recovery and 6 min passive between series	

Table 13.13 HIIT Type 2 (3%), Metabolic (O_2 Plus Neuromuscular), in Developmental Phase 1 of the Preparatory Period in Elite 1500 m Runners With Group 1 Versus Group 3 Profiles

HIIT with long intervals: none HIIT with moderate-duration intervals: none HIIT with short intervals: 106% to 108% of the $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
• 3 (4 × 250 m) at 106% $\dot{V}O_{2\max}$ with recovery (alternating walking and jogging) of 2:15 passive between reps and 6 min passive between series • 4 (300+200+200 m) to 108% $\dot{V}O_{2\max}$ with recovery (alternating walking and jogging) of 2 min between 300 and 200 and then 1:30 between the 200s, and 5 passive min between series	• 3 (4 × 300 m) at 106% of the $\dot{V}O_{2\max}$ with recovery (alternating walking and jogging) of 1:45 passive and 6 min active between series • 3 (6 × 200 m) to 108% of the $\dot{V}O_{2\max}$ with a recovery (alternating stepping and jogging) of 1:30 between reps and 5 min active between series

This work can be carried out uphill (5%-7% slope) by decreasing the quantity by a smaller number of series (two instead of three) or by reducing the number of repetitions per series and by adjusting the pace (about 10% less than the $\dot{V}O_{2\max}$ pace). Examples: 2 × (4 × 300 m) or 3 × (4 × 200 m) uphill. Light jog recovery on descent. More than twice the effort time between repetitions and 6 min walk between runs.

Table 13.14 HIIT Type 3 (2%), Metabolic (O_2 Plus Anaerobic), in Developmental Phase 1 of the Preparatory Period in Elite 1500 m Runners With Group 1 Versus Group 3 Profiles

HIIT with long intervals: paces greater than 102%/104% $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
• 4×800 m in form (600 m to 102% $\dot{V}O_{2\max}$+last 200 m with maximum acceleration), passive recovery of 8 min between repetitions • 5×600 m in form (400 m to 104% $\dot{V}O_{2\max}$ then acceleration over 200 m), recovery of 6 min (walking and jogging) between repetitions	• 3×1200 m in form (400 m to 104% $\dot{V}O_{2\max}$+400 m average pace 102% $\dot{V}O_{2\max}$+400 m to 104% $\dot{V}O_{2\max}$), passive recovery 12 min between repetitions • 3×(1000 m+400 m) at 100% $\dot{V}O_{2\max}$ for 1000 m and 102%/104% $\dot{V}O_{2\max}$ for 400 m with recovery of 2 min jog between 1000 m and 400 m and 8 min between repetitions
HIIT with moderate-duration intervals. Average intervals: paces greater than 110% $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
2×(3×400 m) • First series: first 400 m at 106% $\dot{V}O_{2\max}$, second 400 m faster at 108% $\dot{V}O_{2\max}$, third 400 m at 110% $\dot{V}O_{2\max}$, recovery of 3 min between repetitions and 10 min before the second series • Second series: first 400 m at 110% $\dot{V}O_{2\max}$, second 400 m at 108% $\dot{V}O_{2\max}$, third 400 m pace at 110% $\dot{V}O_{2\max}$, recovery 3 min between 400 m	2×(2×600 m) in form (500 m at 108% $\dot{V}O_{2\max}$ +last 100 m faster), recovery of 6 min between repetitions and 10 min between series
HIIT with short intervals: from 110% to 114% $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
2×(5×200 m) at 112% $\dot{V}O_{2\max}$ for the first three 200 m repetitions and 114% $\dot{V}O_{2\max}$ for the fourth and fifth 200 m repetitions, recovery of 2 min (marching) between repetitions and 8 min between series	6×300 m at 110% $\dot{V}O_{2\max}$ for 200 m then finish faster in the last 100 m, 2 min recovery (stepping) between repetitions

objectives touching type 4 metabolic (anaerobic plus muscle neuro) and type 5 metabolic (peripheral O_2 plus anaerobic and muscle neuro).

For phase 2 (March to mid-May), HIIT distribution moves to 25% with the following breakdown of target types: type 1, 10% (table 13.15); type 2, 5% (table 13.16); type 3, 5% (table 13.17); type 4, 5% (table 13.18); type 5, 0%. The objective of this phase is to maintain the quality developed during the previous months. The breakdown of this phase is detailed in tables 13.15 through 13.18. Note: There are no objectives touching target 5 metabolic (peripheral O_2 plus anaerobic and muscle neuro).

Precompetitive Period (Mid-May to End of June)

The precompetitive period runs from mid-May to the end of June, with the 20% of HIIT work broken into the following distribution of target types: type 1, 0%; type 2, 0%; type 3, 5% (table 13.19); type 4, 10% (table 13.20); type 5, 5% (table 13.21). This breakdown is detailed in tables 13.19 to 13.21. Note:

There are no objectives touching type 1, metabolic O_2 , or type 2, metabolic (O_2 plus neuromuscular).

Competitive Period (June to August)

The competitive period is primarily comprised of races. These races are separated by aerobic-type work, aimed at aiding recovery. The forms of HIIT reappear as part of a new work cycle preparatory to the main objective and sometimes as a small reminder between races if races are separated by sufficient time. The distribution of HIIT work shifts to 15%, with a target distribution as type 1, 0%; type 2, 5% (table 13.22); type 3, 0%; type 4, 5% (table 13.23); type 5, 5% (table 13.24). The breakdown is detailed in tables 13.22 to 13.24. Note: There are no objectives touching type 1 or type 3.

Tapering Period Before Main Competition

The tapering period occurs in the week before major competitions. The aim is to gain freshness. The distribution of HIIT work shifts to 10% with

Table 13.15 HIIT Type 1 (10%), Metabolic O₂, in Developmental Phase 2 of the Preparatory Period in Elite 1500 m Runners With Group 1 Versus Group 3 Profiles

HIIT with long intervals: from 94% to 98% of the $\dot{V}O_{2\max}$ on times from 3 to 6 min. This work can be done both on track and in nature (fields or forest).	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
• 2 × (1500 m + 400 m) at 94% $\dot{V}O_{2\max}$ (1500 m) and 100% $\dot{V}O_{2\max}$ (400 m), recovery of 1:30 (jogging) between 1500 m and 400 m and 1:30 active between series • 3 × 1200 m (400 + 400 + 400 m progressive pace at 94%/95%/96% $\dot{V}O_{2\max}$), 4 min active between series • Fartlek as 2 min, 3 min, 4 min, 4 min, 3 min between 92% and 95% $\dot{V}O_{2\max}$, active recovery of 1 min to 1:30 • 2 × (3 × 800 m) at 98% $\dot{V}O_{2\max}$, recovery of 1:45 (jogging) and 6 min between series by feel	• 2 × (6 min + 3 min) or 2 × (2000 m + 1000 m) to 95% $\dot{V}O_{2\max}$, passive recovery of 1:45 and 4 min (jogging) • 4 × 3 min or 4 × 1000 m to 98% $\dot{V}O_{2\max}$, passive recovery of 1:15 to 1:30 (jogging) • 2000 m + 1000 m + 1000 m at 95% and 96% $\dot{V}O_{2\max}$, recovery of 1:30 (jogging) • 1000 m + 2000 m + 1000 m at 98% and 96% $\dot{V}O_{2\max}$, recovery of 1:45 (jogging)
HIIT with moderate-duration intervals: from 100% to 104% of $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
• 2 × (5 × 400 m) to 104% $\dot{V}O_{2\max}$, recovery of 1 min (jogging) and 6 min between series • 3 × (400 + 500 + 400 m) at 102%/100%/102% $\dot{V}O_{2\max}$, 1:15 recovery (jogging) and 4 min between series • 2 × (600 + 500 + 400 m) at 98%/100%/102% $\dot{V}O_{2\max}$, passive recovery of 1:15 and 5 min (jogging) between series	• 3 × (2 × 600 m) at 100% $\dot{V}O_{2\max}$, recovery of 1 min (jogging) and 4 min between series • 2 × (4 × 500 m) at 102% $\dot{V}O_{2\max}$, recovery of 1:15 (jogging) and 6 min between series • 3 × (500 m + 600 m + 500 m) at 102%/103% $\dot{V}O_{2\max}$, recovery of 1:30 (jogging) and 3 min between series
HIIT with short intervals: 104% $\dot{V}O_{2\max}$	
Both profiles	
• 2 × (6 × 300 m) at 104% $\dot{V}O_{2\max}$, recovery 1:30 (light jogging) and 6 min passive between series • 3 × (5 × 200 m) at 104% $\dot{V}O_{2\max}$, recovery 45 s (light jogging) and 6 min passive between series	

Table 13.16 HIIT Type 2 (5%), Metabolic (O₂ Plus Neuromuscular), in Developmental Phase 2 of the Preparatory Period in Elite 1500 m Runners With Group 1 Versus Group 3 Profiles

HIIT with long intervals: no objective	
HIIT with moderate-duration intervals: no objective	
HIIT with short intervals: 106% to 108% of $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
• 3 × (200 + 300 + 200 m) at 108%/106%/108% $\dot{V}O_{2\max}$, passive recovery of 5 min (after first 200) and 2 min (after 300), alternating walking and jogging, and passive recovery of 6 min between series • 4 × (250 + 200 + 200 m) at 106%/107%/108% $\dot{V}O_{2\max}$, recovery of 4 min (alternating walking and jogging) and passive 6 min between series	• 5 × (200 + 300 + 200 m) at 106%/108% $\dot{V}O_{2\max}$, passive recovery of 1:45 and 4 min active between series • 4 × (300 + 300 + 200) to 108% $\dot{V}O_{2\max}$, passive recovery of 1:30 (after first 300) and 1:15 (after second 300), alternating walking and jogging, and 4 min active recovery between series

This work can be carried out uphill (% of the slope 5% to 7% difference in level) by decreasing the quantity by a smaller number of series (two instead of three) or by reducing the number of repetitions per series and by adjusting the intensity (about 10% less than the $\dot{V}O_{2\max}$ pace). Examples: 2 × (4 × 250 m) or 3 × (3 × 200 m) uphill. Light jog recovery on descent. Greater than twice the effort time between repetitions and 6 min walk between runs.

Table 13.17 HIIT Type 3 (5%), Metabolic (O_2 Plus Anaerobic), in Developmental Phase 2 of the Preparatory Period in Elite 1500 m Runners With Group 1 Versus Group 3 Profiles

HIIT with long intervals: up to 110% $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
3 × (500+400 m) with 500 m to 108% $\dot{V}O_{2\max}$ then after passive recovery of 3 min a 400 m at 110% $\dot{V}O_{2\max}$ and 8 min passive between series	3 × 1000 m in progressive form (500 m + 300 m + 200 m) • 500 m at 106% $\dot{V}O_{2\max}$ • 300 m at 108% $\dot{V}O_{2\max}$ • 200 m at 110% $\dot{V}O_{2\max}$ Recovery passive of 8 min between repetitions
HIIT with moderate-duration intervals: paces greater than 110% $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
2 × (3 × 300 m) • first series: first 300 m at 112% $\dot{V}O_{2\max}$, second 300 m faster at 114% $\dot{V}O_{2\max}$, third 300 m at 112% $\dot{V}O_{2\max}$, passive recovery of 2:30 between repetitions and 8 min passive before second series • second series: first 300 m at 110% $\dot{V}O_{2\max}$, second 300 m at 112% $\dot{V}O_{2\max}$, third 300 m at 114% $\dot{V}O_{2\max}$, passive recovery 2:30 between repetitions	3 × (2 × 500 m) in the following form: first 300 m at 108% $\dot{V}O_{2\max}$ + the last 200 m faster (110% $\dot{V}O_{2\max}$), passive recovery of 4 min between repetitions and passive 8 min between series
HIIT with short intervals: paces greater than 114% $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
3 × (200+250+200 m) at 114% $\dot{V}O_{2\max}$ for the first 200 m then 112% $\dot{V}O_{2\max}$ for the 250 m then 114% $\dot{V}O_{2\max}$ for the 200 m, passive recovery of 2 min between repetitions and 6 min passive between series	2 × (4 × 250 m) at 114% $\dot{V}O_{2\max}$ for the first two 250 m, with progressive acceleration for the third and fourth repetitions, passive recovery of 3 min and 2:30 and 2 min, passive recovery of 10 min between series

Table 13.18 HIIT Type 4 (5%), Anaerobic Plus Neuromuscular, in Developmental Phase 2 of the Preparatory Period in Elite 1500 m Runners With Group 1 Versus Group 3 Profiles

HIIT with long intervals: no objective	
HIIT with moderate-duration intervals: from 108% to 112% $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
• 3 × (300+400+200 m) to 108%/106%/110% $\dot{V}O_{2\max}$, passive recovery of 1:45 between reps and 6 min passive between series • 3 × (400+200+300 m) to 108% $\dot{V}O_{2\max}$, passive recovery of 2 min between repetitions and 6 min passive between series	• 2 × (2 × 600 m) to 108% $\dot{V}O_{2\max}$ with the last 150 m in acceleration, passive recovery of 2:30 between 600 m and 6 active min between series • 2 × (500+400+300 m) to 108%/110%/112% $\dot{V}O_{2\max}$, passive recovery of 2 min between 500 and 400 m and 1:45 between 400 and 300 m and 6 min active recovery between series
HIIT with short intervals from 112% to 115% $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
• 4 × (250+200+150 m) at 112%/114%/115% $\dot{V}O_{2\max}$, recovery of 3 min (passive) between the 250 m and 200 m and 2 min between the 200 m and 150 m, 6 min walk between the series • 3 × (150/200/150/200 m) at 114%/112%/114%/112% $\dot{V}O_{2\max}$, recovery of 2:30 passive between repetitions and 6 min passive between series	• 3 × (4 × 250 m) to 112% $\dot{V}O_{2\max}$ accelerating the last 100 m, recovery of 3 min (passive) between repetitions and 8 min (walk) between series • 3 × (5 × 200 m) at 115% $\dot{V}O_{2\max}$, passive recovery of 2:30 between repetitions and 6 min active between series

Table 13.19 HIIT Type 3 (5%), Metabolic (O_2 Plus Anaerobic), in the Precompetitive Period in Elite 1500 m Runners With Group 1 Versus Group 3 Profiles

HIIT with long intervals: ranges from 108% to 116% $\dot{V}O_{2\max}$ and above	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
• 3 × 800 m in form 600 m to 110% $\dot{V}O_{2\max}$ + 200 m with maximum acceleration, recovery of 8 min between series • 2 × 1000 m in progressive form (600/400) with 600 m to 114% $\dot{V}O_{2\max}$ then after 400 m to 110% $\dot{V}O_{2\max}$, passive recovery of 10 min between the 2 repetitions	• 2 × 1200 m with first 400 m at 108% $\dot{V}O_{2\max}$ + 400 m at 106% $\dot{V}O_{2\max}$ + 400 m at 108% $\dot{V}O_{2\max}$, passive recovery 12 min between series • 2 × 1000 m in progressive form (500/300/200), 500 m at 110% $\dot{V}O_{2\max}$ then 300 m at 112% $\dot{V}O_{2\max}$ then 200 m at 114% $\dot{V}O_{2\max}$, recovery of 3 min (jogging) between the 2 repetitions
HIIT with moderate-duration intervals: paces from 112% to 116% $\dot{V}O_{2\max}$ (and above)	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
2 × (3 × 400 m) • first series: first 400 m at 112% $\dot{V}O_{2\max}$, second 400 m faster at 114% $\dot{V}O_{2\max}$, third 400 m at 112% $\dot{V}O_{2\max}$, passive recovery of 3 min between repetitions and passive 10 min before the second series • second series: first 400 m at 114% $\dot{V}O_{2\max}$, second 400 m at 116% $\dot{V}O_{2\max}$, third 400 m at maximum pace, passive recovery 3 min between repetitions	2 × (2 × 600 m) in the following form: 400 m of the 600 m at 116% $\dot{V}O_{2\max}$ + 200 m faster, passive recovery of 6 min between repetitions and passive 10 min between series
HIIT with short intervals: paces greater than 116% $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
• 2 × (5 × 200 m) at 118% $\dot{V}O_{2\max}$ for the first four 200 m and close to the maximum for the fifth 200 m, passive recovery of 2 min between repetitions and 8 min passive between series • 3 × (200/250/200 m) at 114% $\dot{V}O_{2\max}$ for the first 200 m then 116% $\dot{V}O_{2\max}$ for 250 m then 118% $\dot{V}O_{2\max}$ for 200 m, passive recovery of 2 min between repetitions and 6 min passive between series	• 6 × 300 m at 115% $\dot{V}O_{2\max}$ for 200 m then finish the last 100 m faster, passive recovery of 2 min between repetitions • 2 × (4 × 250 m) at 116% $\dot{V}O_{2\max}$ for the first two 250 m, in progressive accelerations for the third and fourth repetitions, progressive passive recovery of 3 min and 2:30 and 2 min and 10 min passive between series

the distribution as follows: type 1, 0%; type 2, 0%; type 3, 0%; type 4, 5% (table 13.25); type 5, 5% (table 13.26). The breakdown for types 4 and 5 is detailed in tables 13.25 and 13.26. Note: There are no objectives touching type 1, type 2, or type 3.

If needed and if time is available, a type 1 (metabolic O_2) workout can be used as a reminder between competitions:

Fartlek: 3 × (1 min/2 min/1 min) near 98% to 100% $\dot{V}O_{2\max}$, recovery of 1 min (jogging) between repetitions, recovery 2 min (active jogging) between series

Table 13.20 HIIT Type 4 (10%), Anaerobic Plus Muscle Neuro, in the Precompetitive Period in Elite 1500 m Runners With Group 1 Versus Group 3 Profiles

HIIT with long intervals at 110% to 114% $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
3 × (600 m + 400 m) with the 600 m at 112% $\dot{V}O_{2\max}$ and the 400 m at 114% $\dot{V}O_{2\max}$, passive recovery of 3 min between 600 m and 400 m and 8 min passive between series	800 m + 400 m with 800 m at 110% $\dot{V}O_{2\max}$ followed by a passive recovery of 2 min and then 400 m at 114% $\dot{V}O_{2\max}$ followed by a passive recovery of 8 min, and then 600 m + 300 m with 600 m at 112% $\dot{V}O_{2\max}$ followed by a passive recovery of 3 min (jogging) and then 300 m at 116% $\dot{V}O_{2\max}$
HIIT with moderate-duration intervals at 112% to 114% $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
2 × (3 × 400 m) to 112% $\dot{V}O_{2\max}$, passive recovery of 2:30 between 400 m and 8 min passive between series 3 × (3 × 300 m) to 114% $\dot{V}O_{2\max}$, passive recovery of 2 min between 300 m and 6 min passive between series	4 × (400 + 300 m) at 112% $\dot{V}O_{2\max}$ for the 400 m and 114% $\dot{V}O_{2\max}$ for the 300 m, passive recovery of 2 min between the 400 m and 300 m and 6 min active between series 3 × (500 + 300 + 300 m) to 110% $\dot{V}O_{2\max}$ for the 500 m and then 112% $\dot{V}O_{2\max}$ for the first 300 m and 114% $\dot{V}O_{2\max}$ for the second 300 m, passive recovery of 2 min between the 500 m and the 300 m and 1:45 between the 2 × 300 m and 8 min active between series
HIIT with short intervals at 112% to 116% $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
3 × (300 + 300 + 200 m) at 114% $\dot{V}O_{2\max}$ then 115% $\dot{V}O_{2\max}$ and 116% $\dot{V}O_{2\max}$, passive recovery of 2 min between repetitions and 8 min passive between series 4 × (300 + 200 + 150 m) at 114%/116% $\dot{V}O_{2\max}$, passive recovery of 1:45 between repetitions and 6 min passive between series	2 × (4 × 300 m) at 114% $\dot{V}O_{2\max}$, passive recovery of 2 min between repetitions and 6 min active between series 2 × (6 × 200 m) at 116% $\dot{V}O_{2\max}$, passive recovery of 1:45 between repetitions and 8 min passive between series

Table 13.21 HIIT Type 5 (5%), Anaerobic (Peripheral O_2 Plus Anaerobic Plus Muscular), in the Precompetitive Period in Elite 1500 m Runners With Group 1 Versus Group 3 Profiles

HIIT with long intervals: no objective	
HIIT with moderate-duration intervals: pace 108% to 110% $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
3 × (300 + 400 + 300 m) at 110%/108%/110% $\dot{V}O_{2\max}$, passive recovery of 1:45 between repetitions and 8 min passive between series 4 × (400 + 300 + 300 m) to 108% $\dot{V}O_{2\max}$, passive recovery of 2 min between repetitions and 6 min passive between series	3 × (2 × 600 m) at 108% $\dot{V}O_{2\max}$ with the last 150 m in acceleration, passive recovery of 2:30 between repetitions and 6 min active between series 3 × (500 + 400 + 300 m) to 108%/109%/110% $\dot{V}O_{2\max}$, passive recovery of 2 min between 500 and 400 m and 1:45 between 400 and 300 m and 6 min active between series
HIIT with short intervals from 110% to 115% $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
4 × (250 + 200 + 150 m) at 110%/112%/115% $\dot{V}O_{2\max}$, passive recovery of 3 min between the 250 m and 200 m and 2 min passive between the 200 m and 150 m and 8 min passive between series 3 × (150/200/150/200 m) at 114%/112%/114%/112% $\dot{V}O_{2\max}$, recovery of 2:30 between repetitions and 6 min (passive) between series	3 × (4 × 250 m) to 110% $\dot{V}O_{2\max}$ accelerating the last 100 m, passive recovery of 3 min between repetitions and 8 min passive between series 3 × (5 × 200 m) at 115% $\dot{V}O_{2\max}$, passive recovery of 2:30 between repetitions and 6 min active between series

Table 13.22 HIIT Type 2 (5%), Metabolic (O₂ Plus Neuromuscular), in the Competitive Period in Elite 1500 m Runners With Group 1 Versus Group 3 Profiles

HIIT with long intervals: from 108% to 110% of $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
800 m + 200 m then 600 m + 300 m then 500 m + 400 m 800 m + 200 m: 800 m at 108% $\dot{V}O_{2\max}$, recovery 3 min (jogging and walking), then 200 m at 116% $\dot{V}O_{2\max}$, passive recovery of 8 min 600 m + 300 m: 600 m at 110% $\dot{V}O_{2\max}$, passive recovery 2:30, then 300 m at 116% $\dot{V}O_{2\max}$, passive recovery 8 min 500 m + 400 m: 500 m at 110% $\dot{V}O_{2\max}$, passive recovery 3 min, then 400 m at 116% $\dot{V}O_{2\max}$ passive 8 min	4 × 800 m at 108% $\dot{V}O_{2\max}$ on the first 600 m with progressive acceleration over the last 200 m (finish quickly), active recovery of 6 min (jogging and walking) between repetitions
HIIT with moderate-duration intervals: from 110% to 112% of $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
4 × (2 × 400 m) at 110% $\dot{V}O_{2\max}$, passive recovery of 2 min between repetitions and 4 min passive between series 3 × (4 × 250 m) at 112% $\dot{V}O_{2\max}$, passive recovery of 1:45 between repetitions and 6 min passive between series	2 × (3 × 500 m) at 110% $\dot{V}O_{2\max}$, passive recovery of 2:30 between repetitions and 8 min passive between series 2 × (500 + 400 + 400 m) at 110%/112%/112% $\dot{V}O_{2\max}$, recovery of passive 2 min between repetitions and 8 active min between series
HIIT with short intervals: from 112% to 115% of $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
3 × (4 × 200 m) at 116% $\dot{V}O_{2\max}$, passive recovery of 2 min (stepping) between repetitions and 6 min passive between series 4 × (300 m + 200 m) with 300 m at 116% $\dot{V}O_{2\max}$, passive recovery 2 min, then 200 m progressive and fast finish, recovery 8 min passive between series	2 × (3 × 300 m) at 112% $\dot{V}O_{2\max}$ for first and second repetitions, 114% $\dot{V}O_{2\max}$ for third repetition, passive recovery of 2 min between repetitions and 6 min active between series 2 × (4 × 250 m) with first two 250 m at 114% $\dot{V}O_{2\max}$, 115% $\dot{V}O_{2\max}$ for third and fourth repetitions, recovery 3 min (jogging and walking) between repetitions and 8 min active between series

Table 13.23 HIIT Type 4 (5%), Metabolic (O₂ Plus Anaerobic Plus Neuromuscular), in the Competitive Period in Elite 1500 m Runners With Group 1 Versus Group 3 Profiles

HIIT with long intervals: no objective	
HIIT with moderate-duration intervals: at 112% to 116% $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
2 × (300 + 400 + 300 m) at 112%/110%/112% $\dot{V}O_{2\max}$, passive recovery of 2 min between repetitions and 8 min passive between series 3 × (400 + 300 + 200 m) at 112% $\dot{V}O_{2\max}$, passive recovery of 2 min between repetitions and 6 min passive between series	3 × (2 × 600 m) to 112% $\dot{V}O_{2\max}$ with the last 150 m with acceleration, passive recovery of 2:30 between repetitions and 6 min active between series 2 × (500 + 400 + 300 m) to 112%/114%/116% $\dot{V}O_{2\max}$, passive recovery of 2:15 between the 500 and 400 m and 1:45 between the 400 and 300 m and 6 min active recovery between series
HIIT with short intervals of 114% to 118% $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
3 × (250 + 200 + 150 m) at 114%/112%/116% $\dot{V}O_{2\max}$, recovery of 3 min (passive) between 250 m and 200 m and 2 min (passive) between 200 m and 150 m and 6 min passive recovery between series 4 × (200 + 150 + 200 m) at 116%/114%/116% $\dot{V}O_{2\max}$, passive recovery of 2:30 between repetitions and 6 min passive between series	3 × (3 × 250 m) at 114% $\dot{V}O_{2\max}$ accelerating the last 100 m, passive recovery of 3 min between repetitions and 8 min active between series 3 × (4 × 200 m) at 118% $\dot{V}O_{2\max}$, passive recovery of 2 min between repetitions and 8 min active between series

Table 13.24 HIIT Type 5 (5%), Anaerobic (Peripheral O₂ Plus Anaerobic Plus Neuromuscular), in the Competitive Period in Elite 1500 m Runners With Group 1 Versus Group 3 Profiles

HIIT with long intervals: no objective HIIT with moderate-duration intervals: no objective HIIT with short intervals of 114% to 116% of $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
• 6 × 300 m to 114% $\dot{V}O_{2\max}$, passive recovery of 3 min between repetitions • 2 × (4 × 200 m) to 118% $\dot{V}O_{2\max}$, passive recovery of 1:45 between repetitions and 8 min passive between series • 3 × (300 + 200 + 150 m) at 116% $\dot{V}O_{2\max}$ for the 300 m and 200 m, 150 m at maximum speed, passive recovery of 3 min between the 300 m and the 200 m and 2 min between the 200 m and the 150 m, recovery of 8 min passive between series	• 3 × (300 + 300 + 200 m) at 114% $\dot{V}O_{2\max}$ for 300 m and 116% $\dot{V}O_{2\max}$ for 200 m, passive recovery of 2 min between 300 m and 1:45 between 300 m and 200 m and 8 min active between series • 3 × (400 + 300 m) at 112% $\dot{V}O_{2\max}$ for the 400 m and 114% for the 300 m, passive recovery of 2 min between repetitions and 10 min passive between series • 2 × (3 × 400 m) at 114% $\dot{V}O_{2\max}$, passive recovery of 3 min between repetitions and 10 min passive between series

Table 13.25 HIIT Type 4 (5%), Anaerobic Plus Neuromuscular, During the Tapering Period in Elite 1500 m Runners With Group 1 Versus Group 3 Profiles

HIIT with long intervals: no objective HIIT with moderate-duration intervals: paces 116% to 118% $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
• 3 × (2 × 300 m) with 200 m at 118% $\dot{V}O_{2\max}$ + last 100 m in acceleration, passive recovery of 3 min between repetitions and 8 min passive between series • 3 × (300 + 300 + 200 m) at 116% $\dot{V}O_{2\max}$ for the 300 m and 118% $\dot{V}O_{2\max}$ for the 200 m, passive recovery of 3 min between 300 m and 2 min between the 300 m and the 200 m and 10 min passive between series	• 2 × (2 × 500 m) with 300 m at 114% $\dot{V}O_{2\max}$ + last 200 m with progressive acceleration, passive recovery of 3 min between repetitions and 10 m passive in between series • 2 × (3 × 400 m) at 114% $\dot{V}O_{2\max}$ for the first 400 m then 116% $\dot{V}O_{2\max}$ for the second and then 118% $\dot{V}O_{2\max}$ for the third, passive recovery of 4 min between the first and second then 3 min between the second and third 400 m, 12 min passive between series
HIIT with short intervals: 120% maximum speed	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
• 6 × 150 m alternately: first 150 m progressive, second 150 m quick release, third 150 m in 50 m increments, fourth 150 m progressive, fifth 150 m in steps, sixth 150 m at maximum pace, passive recovery of 6 min between repetitions • 10 × 100 m at a speed close to the maximum (95%), passive recovery of 3 to 4 min between repetitions • 3 × (150 + 200 + 150 m) at a speed close to the maximum (95%), passive recovery of 6 min between repetitions and 10 min passive between series	• 2 × (4 × 200 m) at 120% $\dot{V}O_{2\max}$ with a finishing acceleration, passive recovery of 4 min between repetitions and 10 m passive in between series • 3 × (2 × 250 m) close to maximum speed over distance (95%), passive recovery of 6 min between repetitions and 12 m passive in between series • 4 × (200 m + 150 m) at a speed close to maximum for the 200 m (95%) and maximum speed for the 150 m, passive recovery of 6 min between repetitions and 10 min passive between series

Table 13.26 HIIT Type 5 (5%), Metabolic (Peripheral O₂ Plus Anaerobic Plus Neuromuscular), During the Tapering Period in Elite 1500 m Runners With Group 1 Versus Group 3 Profiles

HIIT with long intervals: no objective HIIT with moderate-duration intervals: no objective HIIT with short intervals: from 116% to 118% $\dot{V}O_{2\max}$	
800 m/1500 m profile (Group 1)	1500 m/5000 m profile (Group 3)
• 5 × 300 m to 118% $\dot{V}O_{2\max}$, recovery of 3 min between repetitions • 3 × (3 × 200 m) at 120% $\dot{V}O_{2\max}$, passive recovery of 2 min between repetitions and 8 min active between series • 3 × (250 + 200 + 150 m) at 116% $\dot{V}O_{2\max}$ for the 250 and 200 m, recovery of 2 min between the 250 and 200 m, then passive recovery of 1:30 before the 150 m at maximum speed, recovery of 8 min passive between series	• 2 × (400 + 300 + 200 m) at 116% $\dot{V}O_{2\max}$ for the 400 m and 118% $\dot{V}O_{2\max}$ for the 300 and 200 m, passive recovery of 2 min between 400 and 300 m and 1:45 between 300 and 200 m, recovery of 8 min active between series • 3 × (400 + 300 m) at 114% $\dot{V}O_{2\max}$ for the 400 m and 116% $\dot{V}O_{2\max}$ for the 300 m, passive recovery of 2 min between repetitions and 10 min passive between series • 2 × (3 × 300 m) at 116% $\dot{V}O_{2\max}$, passive recovery of 3 min between repetitions and 10 min passive between series

CASE STUDY

Jean Claude Vollmer

During my career as a club coach, then as a national coach with the French Athletics Federation, I had the chance to coach many runners, work with athletes and their coaches at different moments of my career and I have always learned a lot from them. Competitors I have trained over 1500 m include Christian Grunder (French junior champion), Blandine Bitzner in her debut (selected at the Atlanta Olympics), Guy Nunige (3:36:84 in 1992, France), Frédérique Quentin (selected at the 1996 Olympics, several times French champion), Nadir Bosch (3:32:94 world finalist, selected for 3000 m steeplechase at the Atlanta Olympics, nine French championship titles), Bryan Cantero (3:36:08, two-time champion of France in the 1500 m indoor), Morhad Amdouni (3:34:55, semifinalist at the 2015 World Championships, several times champion of France), and training them from their junior years before their emergence as world-class runners Mehdi Baala (3:28:98 over 1500 m, World silver medalist 2003, bronze medalist at the 2008 Olympic Games) and Bob Tahri (3:32:94 over 1500 m, third at the World Championships in the 3000 m steeplechase and former European repeat-record holder in the 3000 m steeplechase, 8:06:91, 8:02:19, and finally 8:01:18).

My mission as a national MD coach has allowed me to meet many first-class national and international runners and their coaches. Thanks to these meetings, I was able to exchange, observe, and study their approaches to 1500 m training. Performance over 1500 meters is the product of complex parameters, more than any other distance, as described. Each athlete, through his unique profile, will grasp the distance, the confrontation with his opponents, alongside his strengths and weaknesses. Weaknesses in one area may be compensated by a strong point in another, and tactics must be adapted accordingly to expose a competitor's potential weaknesses as well. Each coach has a personal approach to his runner's training, ideas, convictions, and training methods, but to be successful, he must also be able to adapt his training routines to the profile of his athlete. All coaches are familiar with this reality, in that training content applied in one runner may be entirely ineffective for another. The 1500 m race is the aggregation of the highly demanding requirements of the distance together with the unique profile of an individual runner. Forty years of coaching have shaped my beliefs on the determinants of 1500 m performance, which is (to me at least) becoming clearer and clearer: a well-developed aerobic system allows a runner to release the all-important speed component, allowing the runner to access the performance he desires. To reach the highest level, this takes talent, but talent is far from sufficient. It is also necessary to train very hard every day for years to become a world-class runner. Along these lines, I appreciate the quote of Kevin Durant: "Hard work beats talent when talent fails to work hard." For example, unlike Bosch (the runner with the greatest talent I ever worked with), Baala had the chance to start in a club that specialized in middle distance with qualified coaches and many high level partners. This allowed him to be involved in more-structured training, but above all, allowed him to build an extremely solid foundation, a necessary condition needed to be able to gradually increase workload both in terms of the required volume and intensity, elements that remained hidden from Bosch unfortunately.